

Lab Assignment 4

Session tracking with Cookies

Title

Implementation of Session Tracking using Cookies in Servlets

Aim

To develop servlet programs that demonstrate **session tracking using cookies** by storing user information and retrieving it across multiple servlet requests.

Learning Objectives

- Understand the concept of **session tracking**
- Learn how cookies store client-side information
- Create and retrieve cookies using servlets
- Transfer data between servlets using cookies
- Implement real-time examples like **username tracking** and **weather preference storage**
- Handle redirection based on cookie availability

Software Requirements

- JDK 8 or above
- Eclipse / NetBeans / IntelliJ IDE
- Apache Tomcat Server
- Web Browser (Chrome / Edge / Firefox)
- Servlet API Library

4.1. Write a Servlet program to create a cookie that stores the username and display the stored username in another servlet.

Code:

storeCookie.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.Cookie;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/storeCookie")
public class storeCookie extends HttpServlet {
    private static final long serialVersionUID = 1L;
    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
        String name = request.getParameter("username");
        Cookie ck = new Cookie("username", name);
        response.addCookie(ck);
        response.setContentType("text/html");
        PrintWriter p = response.getWriter();
        p.println(
            "<p>Cookies Stored</p><br>"
            + "<a href='displayCookie'> Show Cookies</a>"
        );
    }

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {

        doGet(request, response);
    }
}
```

displayCookie.java

```
import java.io.IOException;
import java.io.PrintWriter;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.Cookie;

@WebServlet("/displayCookie")
public class displayCookie extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
        Cookie[] cookies = request.getCookies();
        String value = null;
        if (cookies != null) {
            for (Cookie cookie : cookies) {
                if ("username".equals(cookie.getName())) {
                    value = cookie.getValue();
                    break;
                }
            }
            response.setContentType("text/html");
            PrintWriter p = response.getWriter();
            p.println("<h1> Welcome "+value+"</h1>");
        }
    }

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
        doGet(request, response);
    }
}

cookieGet.html
<!DOCTYPE html>
<html>
<head>
```

```
<meta charset="UTF-8">
<title>Get Cookies</title>
</head>
<body>
<form action="storeCookie" method="get">
    Username: <input type="text" name = "username"><br><br>
    <input type="submit" value="Submit">
</form>
</body>
</html>
```

Output:

Username:

Cookies Stored

[Show Cookies](#)

Welcome Shriyash

4.2.A weather servlet is a servlet that receives a request for the weather of a particular city from the user. When the user accesses this servlet and if there is a cookie called city on the client side then it should return the value of cookie and display the weather of the city. If the cookie is absent it redirects to an HTML file that displays the name of cities for which weather info can be obtained.

Code:

WeatherServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.Cookie;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/WeatherServlet")
public class WeatherServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        Cookie cookies[] = request.getCookies();

        boolean cityFound = false;
        String cityName = "";

        if(cookies != null) {
            for(int i = 0; i < cookies.length; i++) {

                if(cookies[i].getName().equals("city")) {

                    cityName = cookies[i].getValue();
                    cityFound = true;

                    out.println("<html><body>");
                    out.println("<h2>Weather Report</h2>");
                    out.println("<h3>City: " + cityName + "</h3>");
```

```
        if(cityName.equals("Pune"))
            out.println("Weather: Pleasant, 25°C");
        else if(cityName.equals("Mumbai"))
            out.println("Weather: Humid, 30°C");
        else if(cityName.equals("Delhi"))
            out.println("Weather: Hot, 35°C");

        out.println("</body></html>");

        break;
    }
}

if(!cityFound) {

    response.sendRedirect("cities.html");
}

protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
    doGet(request, response);
}

}
```

SetCityCookie.java

```
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.Cookie;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/SetCityCookie")
public class SetCityCookie extends HttpServlet {
    private static final long serialVersionUID = 1L;
```

```
protected void doGet(HttpServletRequest request, HttpServletResponse response) throws  
ServletException, IOException {
```

```
    String city = request.getParameter("city");
```

```
    Cookie cookie = new Cookie("city", city);  
    response.addCookie(cookie);
```

```
    response.sendRedirect("WeatherServlet");}
```

```
protected void doPost(HttpServletRequest request, HttpServletResponse response)  
throws ServletException, IOException {  
    doGet(request, response);  
}
```

```
}
```

Cities.html

```
<html>
```

```
<head>
```

```
<title>Select City</title>
```

```
</head>
```

```
<body>
```

```
<h2>Select City for Weather Information</h2>
```

```
<form action="SetCityCookie" method="get">
```

```
<select name="city">
```

```
<option value="Pune">Pune</option>
```

```
<option value="Mumbai">Mumbai</option>
```

```
<option value="Delhi">Delhi</option>
```

```
</select>
```

```
<br><br>
```

```
<input type="submit" value="Get Weather">
```

```
</form>
```

```
</body>
```

```
</html>
```

Output:

Select City for Weather Information

Mumbai ▼

Get Weather

Weather Report

City: Mumbai

Weather: Humid, 30°C

Conclusion :

Session tracking using cookies helps maintain user data across multiple requests in servlets. Cookies store information on the client side and allow data sharing between pages, improving user experience. It is simple to implement but should be used carefully due to security and size limitations.